KRRISH BHASIN

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EDUCATION

BITS Pilani, K.K. Birla Goa Campus — B.E. (Hons.) Computer Science *Expected May 2028*

Coursework: Linear Algebra, Probability & Statistics, Foundations of Data Science, Machine Learning, OOP *Goa, India*

- Codeforces Specialist — peak rating **1550**, 790+ problems solved (handle: Levi0p)

EXPERIENCE

ML Research Intern — Indian Institute of Technology Bombay *Sep 2025 – Feb 2026*

NLP / Machine Translation Lab, advised by Prof. Ganesh Ramakrishnan *Mumbai, India*

- Improved a LLaMA-based LLM post-correction pipeline with domain-specific glossary injection, raising BLEU from **0.211 to 0.271 (+28%)** on internal benchmarks for technical and rare-domain vocabulary.
- Fine-tuned IndicTrans2 in Fairseq on 1.2M sentence pairs across 12 Indic language pairs for multilingual machine translation.
- Diagnosed and fixed tokenization mismatches that were destabilizing training runs in the Udaan OCR/MT pipeline for digitizing Indian-language educational content.
- Ran controlled ablations comparing fine-tuned vs zero-shot performance (BLEU, chrF) and presented findings to the lab.

Software Engineering Intern — Hodo Stays *May 2026 – Jul 2026*

LLM-powered guest communication for a hospitality SaaS platform *Bengaluru, India*

- Designed and shipped a multi-channel (chat, email, in-app) guest messaging system that automated routine guest communication.
- Engineered a three-class intent classifier (booking / maintenance / general) reaching **~91%** accuracy, cutting manual message triage.
- Designed prompt templates with dynamic context injection (guest name, property, booking status) to produce brand-aligned, grounded responses.
- Built the backend on Supabase (PostgreSQL, real-time subscriptions) and routed model calls through OpenRouter, maintaining **~1.4s** median response time.

PROJECTS

Improving NER with a Mixture-of-Experts Layer | *Python, PyTorch, HuggingFace* — Github

- Implemented a Mixture-of-Experts layer from scratch (4 expert MLPs, top-2 differentiable gating, 4.2M params) with a residual + LayerNorm connection to prevent representation collapse, and integrated it into a BiLSTM tagger.
- Designed and ran ablation studies on CoNLL-2003: entity-F1 improved from **71.90% to 72.89%** vs. the baseline.

Speculative Decoding for LLM Inference | *Python, PyTorch, HuggingFace* — *Github*

- Implemented speculative decoding (Leviathan et al., 2023) from scratch, including custom KV-cache management for draft/target models, and validated correctness via accept/reject sanity checks.
- Benchmarked on CPU and GPU (T4) across 20 prompts against autoregressive decoding; identified model scale as the key driver of hardware-dependent speedup.

CO-CURRICULAR

AlgoManiax, Competitive Programming Club, BITS Goa — Core Member *Apr 2025 – Present*

- Mentor peers in competitive programming and organize ICPC-format mock contests and team practice sessions.

TECHNICAL SKILLS

Languages: Python, C++, TypeScript, SQL, Bash

ML / AI: PyTorch, TensorFlow, HuggingFace Transformers, Fairseq, LoRA / PEFT, Scikit-learn, NumPy

LLM & NLP: RAG pipelines, FAISS, prompt engineering, IndicTrans2, BLEU / chrF / ROUGE evaluation

Systems & Web: React, Flask, REST APIs, Supabase (PostgreSQL), Git, Linux, OpenCV